

LE MINH BANG

Academic Curriculum Vitae

Ph.D. Student, EECS International Graduate Program, NYCU

lmbang@cit.ctu.edu.vn | bradonleminhbang1@gmail.com | +84 365 406 653 | [linkedin.com/in/brandon-le-vn](https://www.linkedin.com/in/brandon-le-vn)

Hsinchu City, Taiwan | Can Tho City, Vietnam

ACADEMIC PROFILE

I am a Ph.D. student in the EECS International Graduate Program at National Yang Ming Chiao Tung University (NYCU), Taiwan, supervised by Prof. Ying-Dar Lin and supported by the NYCU Elite Ph.D. Scholarship. I completed my M.S. in the same program in June 2026 with a thesis on input injection defense for LLM-based agents. My research centers on cybersecurity for AI, agentic AI security, and machine learning. Before joining NYCU, I graduated as valedictorian in Software Engineering from Can Tho University and was retained as a teaching assistant and researcher. My long-term goal is to develop practical, trustworthy AI systems and contribute to cybersecurity education.

EDUCATION

- National Yang Ming Chiao Tung University (NYCU)** Jul. 2026 – Present
Doctor of Philosophy, Electrical Engineering and Computer Science (EECS IGP) Hsinchu, Taiwan
 - Advisor: Prof. Ying-Dar Lin.
 - Research focus: cybersecurity, agentic AI security, and machine learning.
 - Recipient of the NYCU Elite Ph.D. Scholarship (2026).
- National Yang Ming Chiao Tung University (NYCU)** Sep. 2024 – Jun. 2026
Master of Science, EECS International Graduate Program Hsinchu, Taiwan
 - GPA: 4.28/4.30; Master's thesis grade: 4.30/4.30.
 - Master's thesis: *Detecting and Preventing Input Injection in LLM-based Agents with 3-Layer Filtering*.
 - Advisor: Prof. Ying-Dar Lin.
 - Research focus: AI-driven cybersecurity, LLM-based agents, and network security.
 - Selected coursework: Deep Learning, Data Mining, Computer Vision, Computer Networks, Data Visualization and Visual Analytics.
- Can Tho University (CTU)** Sep. 2018 – Dec. 2022
Bachelor of Engineering in Software Engineering Can Tho, Vietnam
 - GPA: 3.98/4.00; graduated as Valedictorian of the Software Engineering cohort.
 - Undergraduate thesis grade: 10.0/10.0, advised by Dr. Truong Minh Thai.
 - Focus areas: software development, machine learning, information security, and Internet of Things.

RESEARCH AND PROFESSIONAL EXPERIENCE

- High-Speed Networking Lab (HSLab), NYCU** Oct. 2024 – Present
Doctoral Researcher (Jul. 2026 – Present); Graduate Researcher (Oct. 2024 – Jun. 2026) Hsinchu, Taiwan
 - Research LLM-agent security, input injection attacks, RAG/tool-output risks, and practical defense mechanisms for AI-driven cybersecurity systems.
 - Lab leader for hacking competitions and cybersecurity practice, including ASIS CTF 2025, DECONstruCT.F 2025, UMDCTF 2026, and picoCTF training activities for lab members.
- Can Tho University** Dec. 2022 – Present; officially appointed Jun. 2023
Teaching Assistant & Researcher Can Tho, Vietnam
 - Support teaching activities, examinations, laboratory sessions, and student mentoring in the Faculty of Software Engineering.
 - Participate in applied ICT, IoT, microservice, and data-management projects for agriculture and aquaculture with academic, governmental, and international partners.

RESEARCH INTERESTS

- **Cybersecurity for AI:** LLM-agent security, agentic AI security, input injection defense, RAG poisoning, tool-output injection, trustworthy AI systems, and adaptive security mechanisms.
- **Machine Learning:** Deep learning, data mining, and learning-based detection models for security applications.
- **AI Application for IoT:** AI-enabled monitoring, data-driven decision support, environmental sensing, and applied ICT systems for agriculture and aquaculture.

PUBLICATIONS

- [1] Truong Minh Thai, Le Nhat Bang, Tran Le Nhu Y, Huynh Quy Khang, Truong Phuc Vinh, and **Le Minh Bang**. “An Integrated Model of AI Agent for the Aquaculture Sector.”
- [2] Truong Minh Thai, Huynh Quy Khang, Ngo Thi Thu Thao, Luu Ca, and **Le Minh Bang**. “The Microservice Based Architecture for IoT Systems in Agriculture and Aquaculture.”

SELECTED RESEARCH AND DEVELOPMENT PROJECTS

- **Three-Layer Filtering Framework for LLM-Agent Input Injection Defense** 2024 – 2026
Master’s thesis research, NYCU
 - Designed and evaluated a cascaded defense framework for detecting malicious inputs in LLM-based agents across prompt, retrieval, tool-output, and interaction-level contexts.
 - Built experimental pipelines for trace collection, model training, detection, and end-to-end security evaluation under realistic agent workflows.
 - Thesis awarded the maximum grade of 4.30/4.30.
- **Merry Year Project: Salinity-Control Management System** 2022 – 2025
Can Tho University; collaboration with the Government of Hau Giang Province and Merry Year Korea
 - Contribute to the development of a salinity-control management system supporting agricultural decision-making and environmental monitoring in the Mekong Delta.
 - Participate in software design and implementation for data management, monitoring, and practical deployment with local stakeholders.
- **JICA Japan Project Model 11: ICT Software for Agriculture and Aquaculture** 2022 – 2025
Can Tho University; JICA Japan-supported ICT-in-agriculture project
 - Contributed to software development for supporting farming-process management in agriculture and aquaculture.
 - Implemented data management and monitoring functions for rice farming and shrimp farming models based on iterative project requirements.

HONORS AND AWARDS

- **NYCU Elite Ph.D. Scholarship** 2026
National Yang Ming Chiao Tung University
 - Competitive scholarship awarded to outstanding doctoral candidates based on academic record and research potential.
- **TaiwanICDF Scholarship** 2024
Taiwan International Cooperation and Development Fund
 - Fully funded scholarship awarded for graduate study in Taiwan based on academic merit and development potential.
- **Valedictorian and Outstanding Graduate** 2023
Can Tho University
 - Recognized for achieving the highest overall academic performance in the Software Engineering cohort.
- **KMS Back2School Scholarship** 2022
KMS Technology
 - Selected competitively based on academic performance and technical potential.
- **Shine with ADG Scholarship** 2021
ADG Distribution
 - Awarded to high-achieving students in information technology programs.
- **Tran Dai Nghia Scholarship – Talented Student** 3 academic years
Vinh Long Radio – Television Station
 - Awarded for academic excellence and outstanding student performance.
- **Academic Excellence Awards** 3 consecutive academic years
Can Tho University
 - Recognized for excellent academic results within the class.

TEACHING AND MENTORING

- **Teaching Support and Student Mentoring** Dec. 2022 – Present
Faculty of Software Engineering, Can Tho University
 - Assist lecturers in teaching activities, examinations, and laboratory/practical sessions.
 - Mentor students in software engineering, programming practice, and applied technical projects.
- **Cybersecurity Practice and CTF Training** 2024 – Present
High-Speed Networking Lab, NYCU
 - Lead and coordinate lab participation in hacking competitions and CTF-based cybersecurity practice.
 - Support picoCTF training and peer learning through challenge selection, progress tracking, and discussion-oriented practice.

CERTIFICATIONS AND LANGUAGES

- **English:** TOEIC 895/990; B2/strong working proficiency.
- **Vietnamese:** Native.

REFERENCES

1. **Prof. Ying-Dar Lin**
Chair Professor, Department of Computer Science
National Yang Ming Chiao Tung University (NYCU), Taiwan
Email: ydlin@cs.nctu.edu.tw
Relationship: Master's Thesis Advisor and Doctoral Advisor
2. **Dr. Truong Minh Thai**
Dean, Faculty of Software Engineering
College of Information and Communication Technology
Can Tho University (CTU), Vietnam
Email: tmthai@ctu.edu.vn
Relationship: Undergraduate Thesis Advisor and Research Supervisor